

It is all in the [MASK]: Simple instruction-tuning enables BERT-like masked language models as generative classifiers

Benjamin Clavié*, Nathan Cooper, Benjamin Warner

Answer.AI, 2093 Philadelphia Pike, Suite 3112, Claymont, DE 19703, United States of America

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ABSTRACT

While encoder-only models such as BERT and ModernBERT are ubiquitous in real-world NLP applications, their conventional reliance on task-specific classification heads can limit their applicability compared to decoder-based large language models (LLMs). In this work, we introduce ModernBERT-Large-Instruct, a 0.4B-parameter encoder model that leverages its masked language modeling (MLM) head for generative classification. We design a simple approach, extracting all single-token answers from the FLAN dataset collection, and repurposing standard MLM pre-training to only mask this single token answer. Our approach employs an intentionally simple training loop and inference mechanism that requires no heavy pre-processing, heavily engineered prompting, or architectural modifications. ModernBERT-Large-Instruct exhibits strong zero-shot performance on both classification and knowledge-based tasks, outperforming similarly sized LLMs on MMLU and achieving 93% of Llama3-1B's MMLU performance with 60% less parameters. We also demonstrate that, when fine-tuned, the generative approach using the MLM head matches or even surpasses traditional classification-head methods across diverse NLU tasks. This capability emerges specifically in models trained on contemporary, diverse data mixes, with models trained on lower volume, less-diverse data yielding considerably weaker performance. Although preliminary, these results demonstrate the potential of using the original generative masked language modeling head over traditional task-specific heads for downstream tasks. Our work suggests that further exploration into this area is warranted, highlighting many avenues for future improvements.

1. Introduction

Since the release of BERT (Devlin et al., 2019) in 2018, Transformer (Vaswani et al., 2017)-based encoder-only pretrained models have been omnipresent in Natural Language Processing, being used to perform a wide variety of tasks and set state-of-the-art results across many of them. In the following years, much work has focused on further pretrained language models, with increasing emphasis on generative, decoder-only models trained on vast quantities of data, generally called Large Language Models (LLMs). This trend has intensified following the release of ChatGPT and the impressive performance of instruction-tuned AI assistants in general (Lambert et al., 2024; Dubey et al., 2024; QwenTeam et al., 2025). Notably, LLMs have demonstrated their strong potential as zero-shot (Brown et al., 2020) and few-shot (Radford et al., 2019) learners, especially when instruction-tuned (Wei et al., 2021).

However, while LLMs are very powerful, a large number of real-world tasks continue to be performed by encoder-only models, such as BERT, RoBERTa (Liu et al., 2019), or DeBERTaV3 (He et al., 2023a). There are multiple reasons for the continued prevalence of encoder

models. One important factor is their greatly reduced cost; they can process large volumes of data with computational requirements orders of magnitude lower than those of LLMs (Warner et al., 2024b). Moreover, although decoder-only models are often better logical reasoners thanks to their autoregressive nature and techniques such as Chain-of-Thought prompting (Wei et al., 2022), encoder-only models have consistently demonstrated strong performance on more traditional tasks, such as text classification, where fine-tuned encoder-only models can match the performance of LLMs that have 100× more parameters (Li et al., 2022).

When using encoder-only models, classification tasks are traditionally performed with task-specific classification heads, where the model's representations are pooled into a single vector that is then used to assign an appropriate label. However, this approach has traditionally lagged behind in zero-shot classification, as it requires a substantial fine-tuning step to yield good results. As a result, zero-shot classification methods have typically relied on leveraging the classification head by reframing tasks in an appropriate way. The dominant approach has been to repurpose textual entailment: after extensive pretraining on

* Corresponding author.

E-mail addresses: bc@answer.ai (B. Clavié), nc@answer.ai (N. Cooper), bw@answer.ai (B. Warner).

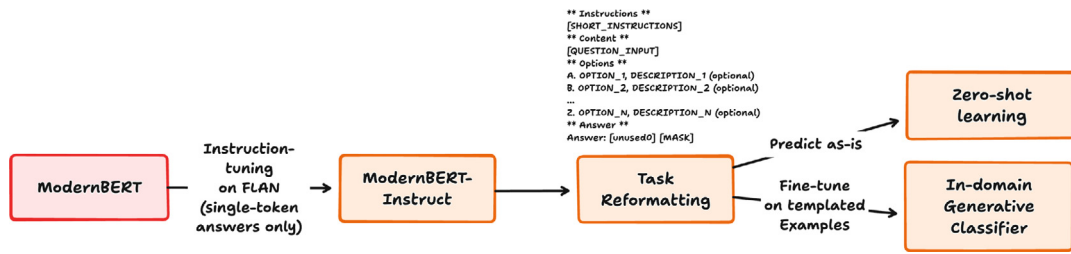


Fig. 1. High level overview of the process.

entailment tasks such as MNLI (Williams et al., 2018), where the model predicts whether a sentence is entailed or contradicted by another, the final model is used to determine whether a specific label is entailed by a given sentence (Lewis et al., 2020). Further studies have shown that framing a very large number of tasks as entailment tasks can produce an even stronger zero-shot classifier (Sileo, 2024). Another promising approach treats zero-shot classification as a form of next-sentence prediction (Liu et al., 2023).

Researchers have explored using the Masked Language Modeling (MLM) head, the generative component of encoder-only models, for classification. Early studies demonstrated strong potential in few-shot learning by turning tasks into Cloze-style questions (Taylor, 1953), where filling in a single token suffices to assign a label. For example, Pattern-Exploitive Training (Schick and Schütze, 2021a) has been used to let models generate probabilities over multiple labels using multiple prompts, showing that these probabilities can transform MLM models into quasi-autoregressive few-shot learners (Schick and Schütze, 2021b) with only a few training steps per task. A recent study further demonstrated that MLM models can be converted into capable autoregressive in-context learner without any additional training by appending multiple [MASK] tokens at the end of the document. However, this effect is only apparent in models with 900 million parameters or more (Samuel, 2024).

Other studies have shown that prompt-based approaches can achieve strong results on classification and entailment tasks. However, these methods require custom prompts for each task and often perform worse than traditional classification-head methods (Sileo, 2024). Moreover, these approaches are highly sensitive to the choice of tokens used to verbalize a label, with performance varying significantly even among semantically similar tokens. To address these issues, UniMC (Yang et al., 2022) introduced an additional step by converting tasks into a multiple-choice format using semantically neutral verbalizers (e.g., “A”, “B”) instead of meaningful words, and by employing a modified attention mask that ensures each verbalizer attends exclusively to its corresponding label description. Although this method achieved particularly strong results across a series of tasks, it adds considerable overhead.

Finally, it is worth noting that encoder-only models, while prevalent in the industry and representing the majority of model downloads from the HuggingFace Hub (Warner et al., 2024b), are predominantly older models. In fact, the most widely used models are the original BERT (Devlin et al., 2019) (from 2018/2019), RoBERTa (Liu et al., 2019) (from 2020), and DeBERTaV3 (He et al., 2023a) (from 2022), each with different strengths and weaknesses. This contrasts sharply with LLMs, where the vast majority of popular models are recent releases (Dubey et al., 2024; QwenTeam et al., 2025; Liu et al., 2024) and are updated multiple times per year (Touvron et al., 2023; Dubey et al., 2024; Yang et al., 2024; QwenTeam et al., 2025). These newer releases come with considerable improvements in many area, particularly in the diversity and quality of data mixes which have been shown to have a large impact on performance in zero- and few-shot tasks (Soldaini et al., 2024; Li et al., 2024; Penedo et al., 2024). The recent release of ModernBERT (Warner et al., 2024a), trained using a modernized architecture and a large-scale, modern data mix, thus prompts further exploration into whether similar gains in zero-shot generative performance can be obtained with encoder-only models.

1.1. Contributions

Many existing approaches to generative classification using encoder-only masked language models rely on substantial overhead. These methods demand either extensive prompt design, require converting the model to operate in an autoregressive manner (thereby reducing the computational efficiency benefits of single-forward pass models), or necessitate adjustments to the attention mechanism.

In this paper, we showcase the potential of a simple generative approach to encoder-only models that requires little-to-no prompt engineering and no significant modifications to the model. We do so by introducing ModernBERT-Large-Instruct, an encoder-only model finetuned to perform classification tasks through its MLM head without further modification (see Fig. 1). We show that ModernBERT-Large-Instruct, although trained on the older FLAN instruction dataset, is remarkably strong for its size. Our approach is built around an extremely simple training loop to highlight its potential without heavy engineering requirements.

Zero-Shot. We show that ModernBERT-Large-Instruct is remarkably strong in multiple application settings. On classification tasks, it is competitive with previous approaches, outperforming all of them in true zero-shot settings on two out of three tasks. On zero-shot MMLU and MMLU-Pro, which evaluate the model’s knowledge and reasoning ability, it is the best overall model of its size class on MMLU, outperforming even similarly sized LLMs such as SmoLLM2 (Allal et al., 2024) and performing closer to the much larger Llama3.2-1B (Dubey et al., 2024). On MMLU-Pro, it is the second-best model of its size class, losing only to another encoder model that employs an attention mask specifically designed to improve multiple-choice question performance.

Fine-Tuning. We also show that once finetuned, ModernBERT-Large-Instruct either outperforms or matches the performance of fully finetuned classification-head methods across a large number of common classification benchmarks spanning multiple domains, including news subject detection, textual entailment, forum post topic identification, and emotion detection, with even greater gains observed on more fine-grained tasks.

Base Model Impact. Finally, we demonstrate that these results are unique to ModernBERT (Warner et al., 2024a). Models based on older architectures, such as RoBERTa-Large (Liu et al., 2019), or those trained on less varied data mixes, such as GTE-en-MLM-Large (Zhang et al., 2024a), show noticeably worse performance. Our results join a growing body of research (Yang et al., 2022; Schick and Schütze, 2021a; Samuel, 2024) and provide a strong basis for further exploration into leveraging the generative capabilities of masked language models. In particular, we believe that the effectiveness of our simple approach demonstrates the strong potential of low-parameter-count encoder-only masked language models.

2. Related work: Prompting masked language models

As briefly discussed above, this work is not the first instance of attempting to leverage encoder-only masked language models, such as BERT or ALBERT, to perform tasks in a generative manner. However, many of the previous approaches rely on complex modifications. In this

section, we provide a high-level overview of the most used previous methods.

Verbalizer Prompting (Xia et al., 2022; Gao et al., 2021) Prompting masked-language model to generate a single verbalizer in order to perform a given task is not new. Indeed, this approach was explored soon after the gain of popularity of pre-trained transformer-based masked language models, such as RoBERTa (Liu et al., 2019). This approach was further explored with ELECTRA (Clark et al., 2020; Xia et al., 2022). However, these approaches were frequently seen to either to require the definition of task-specific formats and hand-crafted verbalizers, with different formats having a huge impact on downstream performance, thus lowering zero-shot potential (Xia et al., 2022). Moreover, while this method had potential, it has not been shown to match full-head fine-tuning in full-shot settings, although they are remarkable few-shot learners (Gao et al., 2021).

PET & iPET (Schick and Schütze, 2021a,b) Pattern-Exploiting Training is a method which attempts to combine prompting with semi-supervised learning. Similarly to our approach, PET reformulates classification tasks into cloze-style questions, where each training instance becomes a natural language sentence with a masked token representing the label, which the model learns to predict. However, despite promising performance, PET continues to heavily rely on task-specific patterns and verbalizers, limiting scalability and adaptability. Iterative PET (iPET) addressed this by incorporating knowledge distillation in a multi-stage process, using models trained on small labeled datasets to generate pseudo-labels for larger unlabeled datasets, thus progressively enhancing performance. Nonetheless, both PET and iPET function better in few-shot settings and involve multiple training stages and significant computational costs compared to simpler prompting methods. Despite these limitations, PET and iPET provided foundational insights and methods critical to the evolution of modern prompt-based fine-tuning, particularly in low-resource scenarios.

BERTs as In-Context Generative Learner (Samuel, 2024) Recently, Samuel (2024) (Samuel, 2024) showed that masked language models can perform zero-shot and few-shot tasks in an auto-regressive manner, by running multiple iterations in a row with multiple [MASK] tokens appended to the text, and only retaining the model's prediction for the first [MASK] token. While promising, this method is considerably different from our suggested approach, as it requires a considerably higher computational cost, a more complex pseudo-log likelihood calculation in order to perform generative sampling, and appears to greatly benefit from larger model sizes. As such, this method is currently in its very promising early stages, and requires further exploration before it can be considered as an alternative to other, considerably less computationally expensive zero-shot methods.

UniMC (Yang et al., 2022) Finally, UniMC is the closest approach to ours. It relies on a similar mechanism, where all tasks it seeks to perform are re-framed as multiple-choice questions, with the model choosing one of the options in order to perform the task. The main difference between our work and UniMC is in simplicity: despite its strong showings across a large number of tasks, UniMC has known limited real-world adoption, with weaker but easier to use zero-shot approaches such as BART-MNLI (Lewis et al., 2020) being much more widely used. This is largely as UniMC requires the use of complex attention masks in order for each proposed answer to only attend to itself during the model processing. While this method has merits and appears to reach very strong empirical results, it makes large-scale adoption more challenging, as it requires careful additional processing and complicates the approach's compatibility with more modern efficient attention implementations such as Flash Attention (Dao et al., 2022). However, the conceptual underlying idea of UniMC is highly similar to that of this paper, and our work can be thought of an attempt at a simplification and generalization of the approach introduced by UniMC.

3. Methods

3.1. Base model

We theorize that the ability to function as a generative multitask learner might be an emergent capability more modern, larger-scale data mixes, which have been shown to greatly improve zero-shot performance in decoder-only models (Jawaheripi et al., 2023; Soldaini et al., 2024). Another potential factor is the use of a more modern model architecture, more closely resembling the Transformer++ (Touvron et al., 2023; Gu and Dao, 2023) than the original Transformer (Vaswani et al., 2017). Moreover, a model with a context length exceeding 512 tokens, the norm for most existing masked language models (Devlin et al., 2019; Liu et al., 2019), allows for longer instructions and would therefore also be beneficial. We also believe that model size is a strong contributor in unlocking instruction-following, with Flan-T5 showing that substantial gains from multi-task instruction tuning are obtained at larger model sizes, with the smallest model variants showing much more modest improvements (Chung et al., 2024).

We choose to use the recently released ModernBERT (Warner et al., 2024a), which satisfies all of the above requirements, as it is an 8k context-length model using a modernized architecture and trained on a large-scale, modern data mix. We further explore this decision in Section 6 by comparing ModernBERT to other encoders, which do not satisfy all three constraints. Similarly to previous studies, we focus our explorations on ModernBERT-Large, the 395 million parameter ModernBERT variant, as is common practice in the exploration of encoder-only model zero-shot capabilities (Yang et al., 2022; Liu et al., 2023).

3.2. Training setting

We propose a simple training objective to further pretrain a masked language model (MLM) to turn it into a capable zero-shot model. This training objective is a variant of the model's pretraining objective, utilizing the model's MLM head.

Normal MLM In classical MLM training, used to pretrain backbone models such as BERT (Devlin et al., 2019), RoBERTa (Liu et al., 2019), DeBERTa (He et al., 2023b) or ModernBERT (Warner et al., 2024a), a proportion of input tokens, between 15 and 40%, is randomly masked: that is, replaced by a [MASK] token. The model's training objective then becomes a denoising objective: in a single pass, it must predict what the masked tokens originally were.

Masked Instruction Tuning The aim of this work is to functionally instruct-tune the Masked Language Modeling head of an MLM model to use its generative capabilities to perform a wide array of downstream tasks, in a way similar to sequence-to-sequence (Sutskever et al., 2014) models such as T5 (Raffel et al., 2020) and its instruct-tuned variant, Flan-T5 (Chung et al., 2024). An immediate limitation of this method is that, unlike causal language modeling, masked language modeling generates its outputs in a single forward pass, replacing all [MASK] tokens simultaneously. Consequently, we must format our data so that the model is expected to predict only a single token for a given task.

Answer Token Prediction We propose a simple variant, inspired by large language models' instruction tuning pipeline: answer token prediction. This objective is a very simple tweak to the normal MLM objective. Rather than masking multiple tokens throughout the input text, we mask a single token, which is the **verbalizer** for a label or answer. This can be considered a restricted form of sequence-to-sequence learning (Sutskever et al., 2014), akin to the way generative models such as LLMs perform tasks. Effectively, this means that all tasks are reframed in the format of a **Cloze** question, where the input is formatted so answering the question requires generating a single verbalizer token. This templating process is further detailed in Section 3.5.

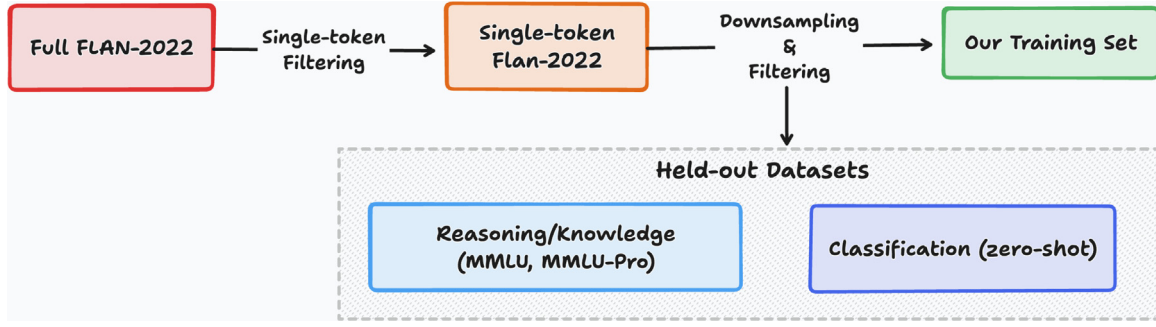


Fig. 2. High level overview of our data processing pipeline.

Verbalizers In the encoder-prompting literature (Xia et al., 2022), a verbalizer refers to a single token that a model outputs to represent its entire answer. Such a token is considered semantically meaningful if it accurately describes the label (for example, the token “Positive” in the context of emotion classification). However, in many cases, it is not possible to express the correct model output with a single meaningful token. In these cases, semantically empty verbalizers, such as “A”, “B”, “C”, “D”, etc., are used to identify potential answers.

Hardware All experiments in this paper are ran on relatively modest hardware. Both evaluation and training runs were conducted using four NVIDIA RTX 4090, with the training wall clock time of our final run taking less than 24 h.

3.3. Training data

The Single Token Constraint Since MLM models are not pretrained in a generative fashion but rather to fill all [MASK] tokens, adding multiple [MASK] tokens could bias the model to consistently predict the longest possible answer. Rather than train the model to ignore extra masked tokens, a setting we leave for future work, we elected to reformulate the instruction tuning data and downstream tasks to require a single [MASK] token response.

Modern Instruction Sets This severely constrains our data selection process, as many of the newer instruction-tuning sets, such as TuluV3 (Lambert et al., 2024), are explicitly constructed to train helpful instruct-tuned assistants. The expected model outputs are noticeably longer than a single token. As a result, many of the most popular instruction-tuning datasets are not suitable for this exploratory work, as they would require considerable pre-processing to be usable.

FLAN Older instruction-tuning datasets which predate the spread of assistant-style models ushered in by ChatGPT, such as the FLAN-2022 (Wei et al., 2021; Longpre et al., 2023) collection, contain a large proportion of examples where the expected model answer is a single token. FLAN is a large-scale instruction-tuning dataset which contains hundreds of datasets further divided into thousands of tasks, with each task’s associated dataset preprocessed to match a specific template. This multi-task fine-tuning data has previously been shown to allow sequence-to-sequence models to reach improved downstream performance across all tasks, including previously unseen tasks (Chung et al., 2024).

Filtering - Data Leakage The FLAN authors originally filtered MMLU (Hendrycks et al., 2021) and BBH (Suzgun et al., 2022) from the data, to serve as held-out test sets. We follow their example and additionally filter out certain classification datasets which we use as downstream evaluation tasks, as detailed in Section 4.1.

Filtering - Downsampling The FLAN collection, in total, contains 396 million examples, of which 120 million are single-token answers. However, a large proportion of those examples come from just a handful of large datasets. The traditional approach to training on FLAN involves downsampling the data (Chung et al., 2024) to cap the number of examples from a single dataset. Following the filtering and downsampling step, our final training dataset contains **20 million examples**. A high-level overview of this filtering process is presented in Fig. 2.

Table 1

Comparison of MMLU results between the three training variants. 100% ATP refers to using Answer Token Prediction as the only training objective. 80% ATP/20% MLM refers to using the ATP objective for 80% of samples and the normal MLM objective for the remaining 20%. 80% ATP/20% dummy refers to using the ATP objective for 80% of samples and dummy MLM examples for the remaining 20%.

Objective	MMLU	MMLU-Pro	Avg
ATP	37.07	13.15	25.11
0.8 ATP/0.2 MLM	41.83	14.74	28.29
0.8 ATP/0.2 dummy	43.06	17.16	30.11

3.4. The elephant in the [MASK]: Dummy examples improve performance

Objective Mix One of the worries with this approach is the potential for catastrophic forgetting due to the relative over-representation of verbalizers with no strong semantic meaning. To attempt to mitigate this effect, we experimented with introducing an Answer Token Prediction vs MLM ratio. Instead of applying answer token masking to all samples, we replace Answer Token Prediction on a subset of samples with standard MLM masking across all tokens. This results in each sample having either a single mask token for the answer, or thirty percent of all tokens randomly masked as during pretraining (Warner et al., 2024a). Based on ablation runs, we choose a Answer Token Prediction vs MLM ratio of 20%,¹ meaning that 80% of our examples used a single masked token objective, as described above, and the remaining examples used the typical random masking MLM pretraining objective.

Dummy Examples However, we introduced a labeling bug where all the true [MASK] tokens were incorrectly labeled as [MASK], rather than as their actual token. This meant that 20% of the predicted tokens were of little value, as the model rapidly learned that in cases where multiple [MASK] tokens are present, their label is always the same, and the loss for these examples plummeted and stayed low throughout training. While we corrected this in a subsequent run, we decided to evaluate all three approaches: only answer token prediction, answer token prediction coupled with 20% MLM, and answer token prediction coupled with 20% “dummy MLM” variant.

Improved Performance We present the performance comparison results in Table 1. We were surprised to discover that the dummy MLM-enriched variant consistently and noticeably outperformed the other methods. To rule out randomness, we replicated this result across multiple seeds. We hypothesize that this effect is due to the dummy examples acting as a form of regularization, perhaps in a way similar to dropout (Srivastava et al., 2014). We believe that this *label dropout* mechanism warrants further investigation and leave it to future work as it is out of the scope of the current paper.

¹ 15% and 20% MLM ratio performed similarly, with a slight edge for 20%. Noticeable degradation was observed with ratios of 10% and lower or 25% and higher.

Final Objective Choice Based on the improved performance we have observed, all pretraining experiments described in this paper use the training objective variant where 80% of the training examples are answer token prediction. The remaining 20% of training samples constitute dummy examples, where tokens are masked as they would be in normal MLM pretraining, but given a single, meaningless label.

3.5. Templating

In order to leverage the Masked Language Modeling head for various tasks, we employ light templating. This templating process is different for training and inference. At training time, our goal is to maximize the number of template variations seen, while at inference time we use simple templates that require minimal modification for new tasks.

3.5.1. Training

During instruction training we add little additional templating to FLAN. Instead, we choose to reuse the extensive templating work already present in the FLAN dataset, based on the assumption that a diverse way of presenting similar tasks will lead to stronger generalization in the final model.

Accordingly, we apply a single templating step to all training examples: we prepend the [MASK] token (which represents the expected answer) with a previously untrained token that serves as an anchor,² indicating that the following token is the model’s final prediction. This is inspired by the use of prefix tokens in the neural information retrieval literature to help the model distinguish between queries and corpus documents (Khatab and Zaharia, 2020; Wang et al., 2022; Zhang et al., 2024b).

Example Training Instance

Frühere Versionen von KDE erlaubten mit einem Trick die Verwendung der Windows®-Taste als Modifizierer (so dass Kurzbe-
fehle wie Win+R möglich waren) und als reguläre Taste (dass mit
Betätigen der Taste Win das K-Menü geöffnet wurde).

Which language is this? [unused0] [MASK]

Above is a randomly sample training example, with the [UNUSED0] anchor token in orange and the masked answer in red.

3.5.2. Inference

During inference, we apply a simple formatting template, with minor tweaks depending on the task. The inference template follows a simple sequence, where (1) basic instructions are given to the model, followed by (2) the text to be labeled or responded to, (3) the list of potential labels, and finally, (4) an answer section that ends in a [MASK] token for the model to predict.

Example Inference Instance (MMLU Hendrycks et al. (2021))

You will be given a question as well as a list of options. Read the question carefully and select the right answer from the list.

QUESTION: An expected side effect of creatine supplementation is:

CHOICES:

- A: muscle weakness.
- B: gain in body mass.
- C: muscle cramps.
- D: loss of electrolytes.

ANSWER:

Answer: [unused0] [MASK]

4. Zero-shot performance

4.1. Setting

Commonsense Reasoning and Knowledge We use two standard LLM benchmarks to evaluate the knowledge and commonsense reasoning capabilities of ModernBERT-Large-Instruct: MMLU (Hendrycks et al., 2021) and MMLU-Pro (Wang et al., 2024). In MMLU, models are asked general knowledge questions across 57 subjects and must select the correct answer out of 4 possible options. To further evaluate the models’ robustness on this task, we use MMLU-Pro, a smaller evaluation set designed to be more challenging. In MMLU-Pro, all questions were filtered to remove trivial knowledge tasks and favor reasoning-heavy questions. Additionally, rather than 4 options, each question has 10, lowering the impact of random guesses and assessing the ability to distinguish among multiple similar answers.

Zero-Shot Classification Evaluating zero-shot classifiers is surprisingly hard, as a large proportion of common benchmarks have been in use for many years, and data contamination during the post-training phase is considerably harder to avoid. The Real-world Annotated Few-shot Tasks (RAFT) benchmark (Alex et al., 2021) contains multiple tasks which are part of FLAN and other commonly used datasets. Its full evaluation process relies on a held-out evaluation mechanism via an online platform which is currently offline. As part of our evaluation, we choose three of the RAFT tasks. Two of them are publicly available and commonly used outside the RAFT suite itself, while the final one is the NeurIPS Impact Statement (NIS) task, where the model must determine if an impact statement involves AI safety risks or not. For this task, we use the annotated training set as our evaluation set.³ The publicly available datasets are ADev2 (Gurulingappa et al., 2012), a medical domain task where the model is given a sentence and must determine whether it is related to a negative reaction to a medication or not; and One Stop English (Vajjala and Lučić, 2018) (OSE), a general domain reading comprehension task where experts rewrote a single sentence into three levels of grammatical complexity: elementary, intermediate and advanced, and in which the model must accurately guess the complexity level. To the best of our knowledge, this setup ensures that only a single task is in-domain for the evaluated models.

Baseline - Encoders We compare ModernBERT-Large-Instruct against three types of baselines. For all tasks, we evaluate other encoder-based zero-shot methods, which are the most commonly used

² We use the [UNUSED0] token present in the ModernBERT vocabulary and left purposefully untrained to support this sort of application.

³ This results in a limited evaluation set, of only 50 examples. However, for this task, the test set itself is only 150 examples; as a result, we believe this sample is an acceptable proxy and a better alternative to data contamination present in most of the other datasets.

Table 2

Result for the models across the zero-shot tasks. Results in **bold** are the best zero-shot results for the under-0.5B parameter model class. Results in *italics* are results where the model is in-domain rather than zero-shot, reported for thoroughness. Results in *italicized bold* are the best results in the 1B-and-above parameter model class. - indicates that the model was not evaluated on this task. Param count ranges are indicated on the left. Results marked with a † are directly reported from the results of Ben Allal et al. (Allal et al., 2024) and results marked with a ‡ are directly reported from the MMLU-Pro Leaderboard (Wang et al., 2024).

		MMLU	MMLU-Pro	ADEv2	NIS	OSE	Average
0.3-0.5B	Tasksource-NLI	36.08	16.54	65.17	58.72	21.11	39.52
	RoBERTa-Large-SST	31.3	13.63	43.61	75.0	40.67	40.84
	UniMC	38.48	18.83	23.29	73.96	36.88	38.29
	ModernBERT-Large-Instruct	43.06	17.16	53.31	85.53	20.62	43.94
	SmoLM2-360M	35.8†	11.38‡	–	–	–	–
	Qwen2-0.5B	33.7†	15.93‡	–	–	–	–
1B+	Llama3.2-1B	45.83	22.6‡	–	–	–	–
	SmoLM2-1.7B	48.44	18.31‡	–	–	–	–
	Qwen2.5-1.5B	59.67	32.1‡	–	–	–	–

low-parameter-count methods for zero-shot applications. These baselines are UniMC (Yang et al., 2022) and Self-Supervised-Tuned RoBERTa-Large (Liu et al., 2023). We also report the best-performing entailment-based method, ModernBERT-Large-Tasksource-nli (Sileo, 2024), as a final baseline, in order to represent this popular family of methods. However, it is worth noting that this model is in-domain on two out of the three classification tasks since zero-shot entailment-based methods are trained on large corpora, encompassing hundreds of tasks, including the vast majority of publicly available classification tasks. Neither method based on Tasksource (Sileo, 2024), a comprehensive collection of tasks rephrased in a format compatible with entailment training, or other popular approaches such as ModernBERT-large-zeroshot-v2.0⁴ are truly out-of-domain on common benchmarks.

Baseline - LLMs We also evaluate low-parameter-count Large Language Models on the commonsense knowledge and reasoning tasks, but exclude them from evaluation on classification tasks due to their overall poor instruction-following ability and their tendency to produce outputs that do not match the classification constraints. On MMLU and MMLU-Pro, we report two classes of LLM results. The first consists of the similar sized SmoLM2-360M (Allal et al., 2024) and Qwen2-0.5B (QwenTeam et al., 2025) models, providing a direct comparison. The second consists of their larger variants, Llama3.2-1B (Dubey et al., 2024), SmoLM2-1.7B (Allal et al., 2024) and Qwen2.5-1.5B (QwenTeam et al., 2025), models that are almost four times larger than ModernBERT-Large-Instruct, to present upper bounds. Whenever possible, we report the results obtained by the authors of SmoLM2 (Allal et al., 2024) for MMLU. This covers all models except Qwen2.5-1.5B and Llama3.2-1B, which we obtained by using the Language Model Evaluation Harness (Gao et al., 2024). For MMLU-Pro, we report results from its official leaderboard.⁵

4.2. Results

Results for the zero-shot experiments are presented in Table 2.

MMLU While non-causal models are often thought to memorize less information than their causal counterparts (Samuel, 2024), our results demonstrate the opposite: on MMLU, ModernBERT-Large-Instruct outperforms all similarly sized models, including both decoder and encoder baselines, with its results being closer to a halfway point between models of its own size and models in the next size class, which have around 4 times more parameters.

MMLU-Pro While ModernBERT-Large-Instruct is the second-best performing on the harder MMLU-Pro, the best-performing approach at the sub-1B scale is also an encoder-only model, with UniMC showing remarkably strong performance. We theorize that UniMC outperforms

ModernBERT-Large-Instruct on MMLU-Pro due to its custom attention masks, ensuring the model represents all labels individually rather than with noise from neighboring labels, while our simple approach forces the model to process the entire input at once with all tokens attending to each other. However, this confirms our overall hypothesis: encoder-only models’ MLM heads appear particularly well suited to zero-shot tasks, even at small model sizes.

Classification On ADEv2 and NeurIPS Impact Statements, ModernBERT-Large-Instruct beats all other zero-shot approaches,⁶ indicating strong performance across domains. However, on One Stop English, it noticeably lags behind two of the other methods, UniMC and Self-Supervised-Tuned (SST) RoBERTa. Both of these models leverage mechanisms not based on a typical classification head, with UniMC leveraging the model’s generative head coupled with custom attention masks, and SST employing a next-sentence prediction objective.

Overview While ModernBERT-Large-Instruct reaches the best zero-shot performance across the evaluated tasks on average, in both classification and knowledge-based QA, it appears to have clear weaknesses, as discussed above. Noticeably, these results point to a clear pattern indicating that various encoder-based methods have different strengths, as evidenced by the large swings in performance across datasets. Interestingly, UniMC (Yang et al., 2022), another MLM-head based classification method, although relying on more complex mechanisms, strongly outperforms ModernBERT-Large-Instruct on two datasets, being the only model to do so, while being noticeably behind on three others. These results further reinforce the strong potential of using “non-traditional” classification heads for zero-shot tasks with encoder models, and warrant further research into the best ways to combine these methods.

5. Full-finetune: Can one head do it all?

5.1. Setting

Having observed the strong zero-shot performance of ModernBERT-Large-Instruct, we now seek to explore whether this behavior holds true in fully fine-tuned settings. The common approach to encoder-model classification is to fine-tune a task-specific head, relying on pooling and a final layer projecting the pooled representation to a hidden dimension matching the number of potential labels.

To do so, we further finetune the ModernBERT-Large-Instruct model resulting from our experiments in Section 3 on a variety of downstream tasks. We use a simple fine-tuning mechanisms where all examples are trained on the single token prediction objective described in Section 3.2.⁷

⁴ <https://huggingface.co/MoritzLaurer/ModernBERT-large-zeroshot-v2.0>

⁵ <https://huggingface.co/spaces/TIGER-Lab/MMLU-Pro>

⁶ Tasksource-NLI’s training set includes ADEv2, meaning it is fully in-domain on this task.

Table 3

Results comparing our approach with a traditional classification head when fully fine-tuned on downstream NLU tasks. For SST-2 and MNLI, marked with a †, we report the results of Warner et al. (Warner et al., 2024a).

	MNLI	Yahoo!	20ng	AGNews	SST-2	IMDB	SST-5	Average
ModernBERT (cls head)	90.8 [†]	77.75	73.96	95.34	97.1[†]	96.52	59.28	84.39
ModernBERT-Large-Instruct	91.03	77.88	73.96	95.24	96.22	97.2	61.13	84.67

Table 4

Downstream results of using different backbone with the same instruction-tuning process.

Model Backbone	MMLU	MMLU-Pro	ADEv2	NIS	Average
ModernBERT-Large	43.06	17.16	53.31	85.53	47.8
GTE-en-MLM-Large	36.69	14.47	20.26	79.48	35.44
RoBERTa-Large	33.11	14.47	16.24	44.0	26.44

Data For this section, we evaluate the models on an array of widely used classification and Natural Language Understanding (NLU) tasks. We select 6 widely used (Liu et al., 2023; Yang et al., 2022; Li et al., 2022) classification benchmarks, comprising of 3 topic classification detection and 3 emotion classification tasks. The three topic classification tasks are AGNews, where a news tagline must be classified into one of 4 categories, Yahoo! Answers Topic (Zhang et al., 2015), a large-scale dataset where messages from the Yahoo Answer forum must be classified into one of ten topics, and 20newsgroup, where posts from online newsgroups must be classified into one of 20 forum categories. The emotion detection tasks are IMDB (Maas et al., 2011), SST-2 and SST-5 (Liu et al., 2019), where movie reviews or extracts from such reviews must be classified into either positive or negative, in the case of IMDB and SST-2, or a finer-grained 5-level classification ranging from Very Negative to Very Positive for SST-5. We also evaluate the model on MNLI (Williams et al., 2018), an extremely common benchmark of NLU capabilities (Wang et al., 2018), where the model is expected to predict the textual entailment between two sentences: that is, whether a hypothesis is entailed by, neutral toward, or contradicted by a given premise.

Baseline We compare a fully fine-tuned, in-domain, ModernBERT-Large-Instruct to a fully fine-tuned, in-domain, ModernBERT-Large, to provide the fairest comparison possible. The models are not evaluated against any external baseline, as the purpose of this task is to evaluate how well ModernBERT-Large-Instruct performs in this setting compared to the more common classification-head based approach. No extensive hyperparameter sweep is conducted; all results reported are the results of a minimal sweep across epoch counts [1,2,3] and learning rates [2e-5, 3e-5, 5e-5] for the MLM setting and [3e-5, 5e-5, 8e-5] for the traditional classification approach. No prompt engineering is performed to keep the comparison as fair as possible.

5.2. Result

The results are presented in Table 3. Overall, it appears that with full-dataset fine-tuning, ModernBERT-Large-Instruct outperforms a fully fine-tuned ModernBERT-Large using a classification head on average. On a task-by-task basis, both approaches trade blows, with both methods coming slightly ahead on certain datasets. Given our extremely simple training loop and limited optimization, we believe that this highlights the robust performance and strong potential of MLM-head based downstream tasks when compared to classification-head based approaches.

⁷ In our limited experiments, it appears that injecting dummy MLM examples, as presented in Section 3.4, helped to improve the pretrained model’s generalization ability but does not improve performance on task-specific fine-tuning, however more exploration of this phenomenon remains warranted.

While the traditional classification method performs better on simpler tasks, such as the SST-2 binary sentiment classification dataset, ModernBERT-Large-Instruct achieves noticeably better results on its more fine-grained version, SST-5, where the emotion expressed in a sentence is broken down into five levels rather than just two. ModernBERT-Large-Instruct’s MNLI performance also highlights that this method is not restricted to single-sentence tasks, but is able to correctly label the relationship between two sentences.

6. Are older encoders also generative classifiers?

A final question we seek to answer, raised in Section 3.1, is whether or not this behavior is specific to ModernBERT, and thus is emergent to the combination of a more modern architecture along with a sizeable mixed-sources 2 billion token pretraining, or was already present in previous encoder models.

6.1. Setting

To answer this question, we reproduce the training setting method presented in Section 3 on two other encoder-only models. The first model is RoBERTa-Large (Liu et al., 2019), one of the best performing and currently the most widely used MLM model.⁸ RoBERTa-base falls within the category of “classical” encoder-only pretrained models, using an architecture very similar to the original BERT (Devlin et al., 2019) model, and trained on limited data sources. We also evaluate GTE-en-MLM-Large (Zhang et al., 2024a), which provides an interesting comparison: while it has been trained on a limited data mix, its architecture is modernized and follows the more modern Transformer++ architecture (Touvron et al., 2023; Gu and Dao, 2023).

6.2. Results

We present the outcome of this experiment in Table 4. The results are stark: RoBERTa-Large, despite a similar parameter count to ModernBERT-Large and almost competitive performance in traditional, classification-head based approaches, achieves poor results in all zero-shot contexts when using the MLM head. GTE-en-MLM-Large (Zhang et al., 2024a), with its modern architecture and longer context length, fares better than RoBERTa-Large on all tasks except MMLU-Pro, where they are equal, but remains considerably weaker than ModernBERT-Large-Instruct.

As such, we hypothesize that strong generalization potential from a Masked Language Model’s MLM head is a property relying mainly on a large-scale, varied pretraining data mix, with architecture itself playing a smaller, but still important, overall role.

7. Conclusion

In this work, we explored the potential of encoder-only masked language models (MLMs) as competitive zero-shot and fine-tuned classifiers through their generative MLM heads. We achieve this by designing a very straightforward training recipe and introducing ModernBERT-Large-Instruct. This model is trained on a filtered subset of the FLAN dataset with a simple training objective. Interestingly, we observed that

⁸ as judged by downloads on the HuggingFace hub.

the use of an unconventional training setup, where 20% of the examples are replaced by dummy samples, vastly outperforms more traditional training approaches.

We demonstrated that modern encoder models can achieve remarkably strong classification performance via the use of their masked language modeling head, in a variety of settings. In zero-shot contexts, ModernBERT-Large-Instruct is competitive with previous approaches requiring more complex training or architectural modifications, even outperforming them on some tasks. On zero-shot knowledge benchmarks such as MMLU, it performs closer to the considerably larger Llama-3.2-1B than the similarly sized SmoLLM2-360M.

When fine-tuned for a single specific tasks, this language-modeling-head classification approach reaches stronger overall performance than traditional classification-head fine-tuning on a wide range of downstream domains, encompassing entailment, topic classification, and emotion detection. These early findings suggest that generative classification with fine-tuned encoder models might represent a better alternative to more commonplace classification approaches.

Finally, we show that these capabilities appear to be unique to encoder-only models trained with modern training recipes, requiring both a modern architecture and modern data mixes, such as ModernBERT. Both older models (e.g., RoBERTa) and modern models trained on lower volume, less-diverse data (e.g. GTE-en-MLM) showed significantly weaker performance across all tasks.

While these are early results, our findings join a small but growing body of work showing that encoder-only models can function well as generative multi-task models. We believe that our work, relying on very simple training and inference mechanisms with a model with a very small parameter count by modern standards (0.4B), demonstrates the strong potential of such approaches.

8. Limitations & future work

Newer Instruction Sets Our work focuses on the exploration of FLAN. While FLAN remains a well-constructed dataset with strong previous empirical results, it is an earlier form of instruction tuning. In modern LLM instruction-tuning, FLAN has often been superseded by much broader instruction sets focusing on more diverse tasks and phrasings (Lambert et al., 2024; Lian et al., 2023; Teknium et al., 2024), although many continue to include FLAN as a subset (Lambert et al., 2024; Lian et al., 2023). However, we choose to use FLAN as the more complex data pre-processing methods required to convert modern turn-based instruction datasets into single token data are out-of-scope for this study. As such, FLAN and its preprocessed format making adapting a large subset of it to a single token generation format very straightforward, represents an ideal experimentation dataset.

Few-Shot and In-Context Learning We focus on exploring the potential of ModernBERT-FLAN and, more generally, of leveraging the MLM head for classification tasks in both zero-shot and full fine-tuning contexts, showcasing its superiority over previous methods. However, an area of particular interest is few-shot learning, either via in-context learning (Dong et al., 2024; Samuel, 2024) or sample-efficient fine-tuning (Schick and Schütze, 2021b). We leave this to future work.

QA-Specific Attention Masks In this work, we focus on a simple training loop to highlight the baseline capabilities of ModernBERT-Large-Instruct. However, previous approaches relying on carefully crafted attention masks such as UniMC (Yang et al., 2022) remain very competitive, despite using an older backbone. We believe that our results should encourage further exploration into modernizing UniMC-like approaches, and we intend to explore this in future work.

Single-Token Generation As our approach relies on generating a single token at a time, this imposes huge constraints on model outputs and necessitates the use of single-token verbalizers, which might often be tokens that themselves carry limited semantic information, such as single letters or digits. The current state-of-the-art methods for

MLM-based generation, such as PET (Schick and Schütze, 2021a) or using BERT-like models as auto-regressive language models out-of-the-box (Samuel, 2024) currently work around this limitation by effectively turning the model into an auto-regressive model rather than treating generation as a single forward pass. We believe that future research should explore whether combining these approaches with our method could further increase performance, while mitigating the noticeably heavier computational cost of using the models as auto-regressive generators.

Scaling Our study focuses on a model with 395M parameters, while the existing instruction-tuning literature has shown that larger models benefit even more from such post-training, especially in the case of FLAN (Tay et al., 2023). However, at the time of our work, no 1B+ masked language model with a recent training mixture was available. We believe that, should such a model become available, our method would transfer directly and yield even greater results.

CRedit authorship contribution statement

Benjamin Clavié: Writing – review & editing, Writing – original draft, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Nathan Cooper:** Investigation, Data curation. **Benjamin Warner:** Writing – review & editing, Validation, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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